

SUPPLEMENT MATERIAL OF THE ARTICLE TITLED: Focal spot size effects in Speckle-Based X-ray Imaging: a Geant4 simulation study

1. Reference images comparison

The process of collecting the reference images was carried out in two ways for the complex thorax sample. The first method consists of collecting the images against a background entirely composed of a vacuum, which, in the most realistic scenario, emulates completely removing the sample from the beam path and moving the modulator through 20 different and totally random positions. The second method consists of leaving the entire background as tissue, without lungs and ribs, so that the images are captured against that background.

Figures S1 and S2 show the images obtained from the UMPA algorithm in the scenarios where is acquired with tissue and without tissue, respectively. This demonstrates that when capturing images against a vacuum background, the quality of the resulting image is higher compared to the tissue background.

The images are obtained for the case of the TiC modulator material with addition of the Al filter.

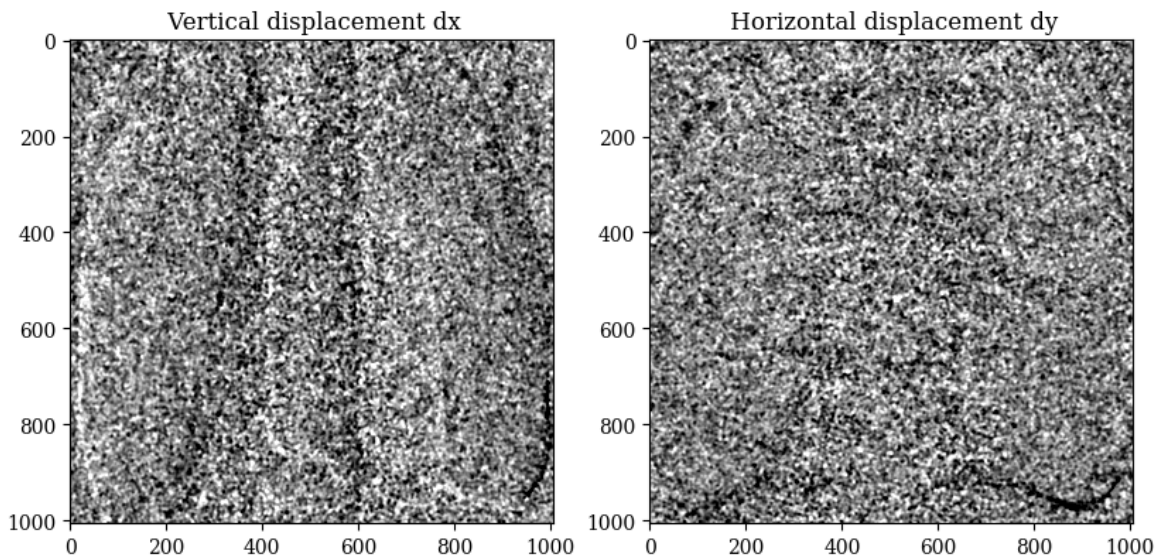


Figure S1. Differential phase maps in the X and Y directions using with tissue as background

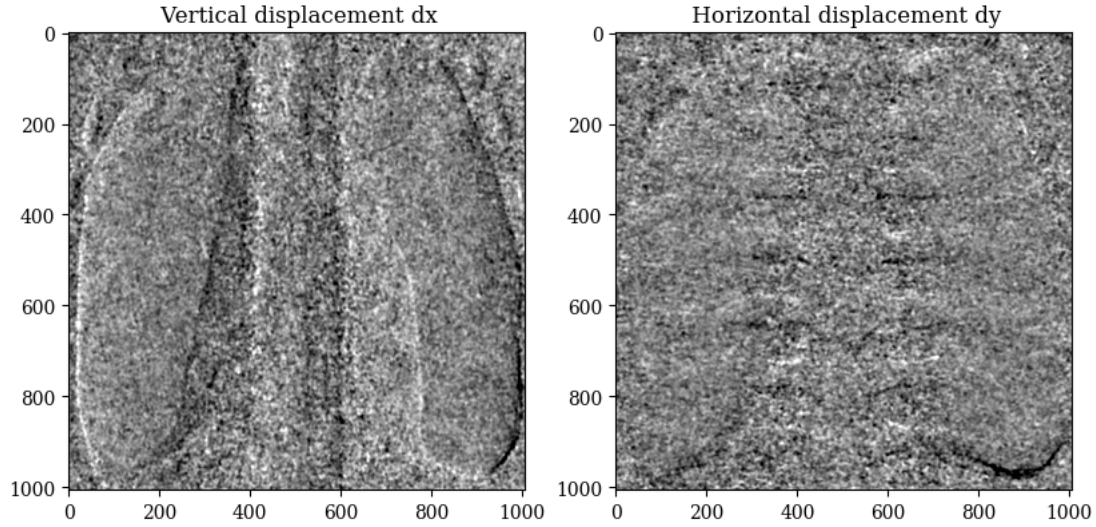


Figure S2. Differential phase maps in the X and Y directions using with vacuum as background.

The same trend is found for the case where the modulator is made of SiC, where it is evident that the signal maps are no longer distinguishable from the background; this trend is mainly attributed to the low visibility that the images present relative to the tissue background.

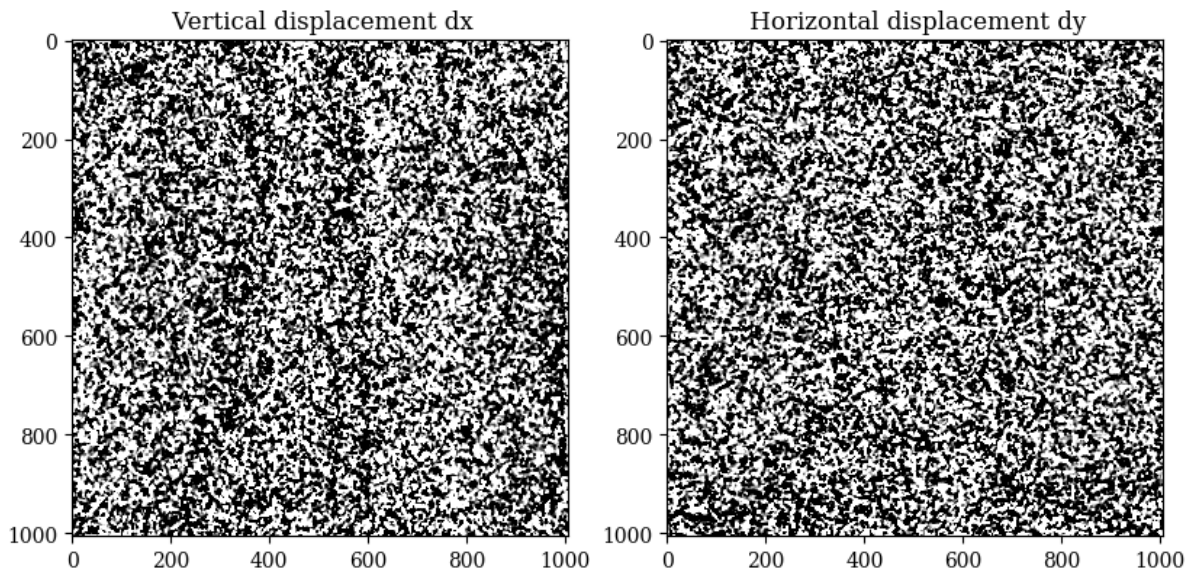


Figure S3. Differential phase maps in the X and Y directions using with tissue as background and SiC as modulator material.

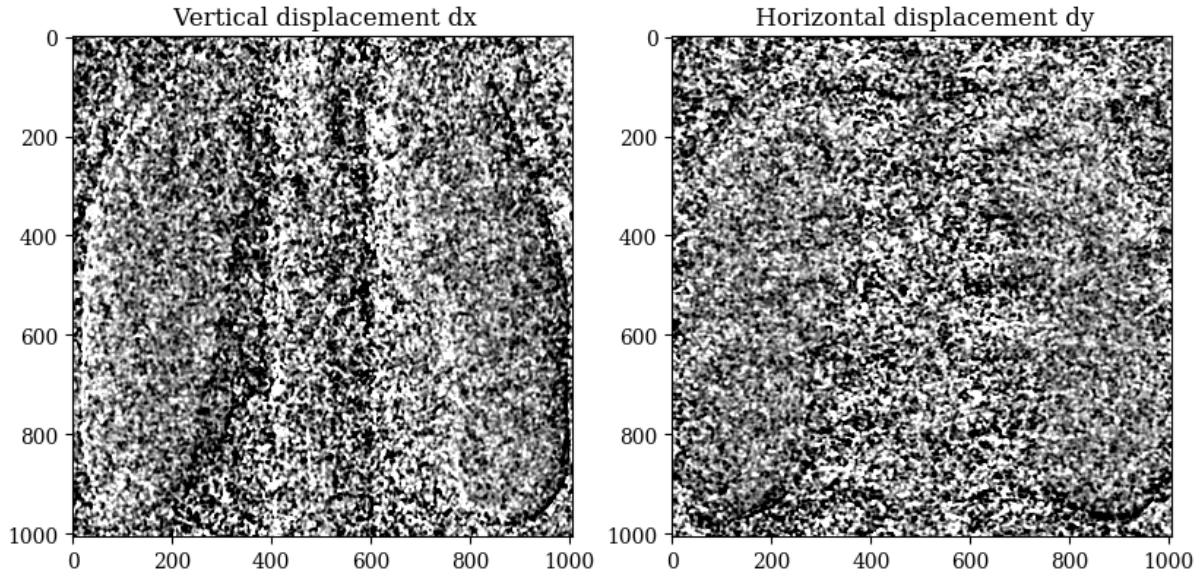


Figure S3. Differential phase maps in the X and Y directions using with vacuum as background and SiC as modulator material.

References

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